

Report on Jenkin Application

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1. Introduction

An online shopping application such as Amazon or Flipkart serves millions of users and is updated frequently with new features, bug fixes, and performance improvements. Manually building, testing, and deploying every code change is slow and error-prone. Continuous Integration and Continuous Deployment (CI/CD) solve this problem by automatically building, testing, and deploying the application every time a developer pushes new code, ensuring that changes reach users quickly and reliably.

2. Jenkins Pipeline

Developer Pushes Code → GitHub → Jenkins Pipeline Triggered → Build → Test → Deploy

- **Build:** Jenkins compiles and packages the application, checking for compilation or build errors.
- **Test:** Automated unit and integration tests run to verify that new changes do not break existing functionality.
- **Deploy:** After all tests pass, Jenkins automatically deploys the application to the target environment.

3. Jenkinsfile (Declarative Pipeline)

```
pipeline {
  agent any
  stages {
    stage('Checkout') {
      steps {
        git branch: 'main',
            url: 'https://github.com/example/online-shopping.git'
      }
    }
    stage('Build') {
      steps { sh 'mvn clean package' }
    }
    stage('Test') {
      steps { sh 'mvn test' }
    }
    stage('Deploy') {
      steps { sh 'sh deploy.sh staging' }
    }
  }
  post {
    success { echo 'Build, test and deployment completed successfully.' }
    failure { echo 'Pipeline failed. Deployment aborted.' }
  }
}
```

4. How Jenkins Ensures Safe Releases

- Every code push to GitHub can automatically trigger the Jenkins pipeline, removing manual steps.
- Build errors are detected early, before faulty code moves further down the pipeline.
- Automated tests catch bugs and regressions before the application is deployed to users.
- Deployment is executed only after the Build and Test stages complete successfully.
- A failed pipeline stops the release immediately, preventing broken code from reaching production.
- Jenkins maintains build history and logs, which help in monitoring and troubleshooting issues.
- Together, these controls make releases faster, repeatable, and significantly safer.

5. Conclusion

Jenkins automates the entire CI/CD process for an online shopping application, from code checkout to build, testing, and deployment. This automation allows new features and bug fixes to be released to users quickly and reliably, while reducing manual errors and minimizing deployment risk. As a result, Jenkins plays a vital role in maintaining a stable and continuously improving application.